Zhu (Leo) Zhi

U.S. Work Authorization: No sponsorship required

Ithaca, NY

zhizhu0730@gmail.com zz766@cornell.edu

linkedin.com/in/zhu-zhi-506499376 github.com/leo-Zhizhu

Education

Cornell University, Ithaca, NY — Expected Graduation in May 2028

Bachelor of Art in Computer Science — GPA: 4.0/4.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Web Development, RSA Cryptosystem, Machine Learning, Linear Algebra, Multivariable Calculus

Technical Skills

Languages: Java, Python, C++, JavaScript, Go, C#, SQL, HTML/CSS

Frameworks & Tools: React, Spring Boot, Node.js, LangChain, LangGraph, Docker, AWS (RDS, ECR, App Runner), ElasticSearch, Git

Systems & Concepts: System Design, CI/CD, OOP, Data Structures, Database Design (SQL/NoSQL), Unit Testing, LLM Systems, Agentic AI, PID Control

Core Strengths: Large-Scale Problem Decomposition, Backend Architecture, ML Model Tuning, Robotics Control Optimization, Cross-Team Communication, Full Stack Development

Work Experience

Ginlix AI — NYC, NY

Full Stack Software Engineer, Product Manager — Dec 2025 – Present

- Built a financial AI assistant platform that handles research, analysis, and code in one place to **provide the most convenient, up-to-date, in depth, and transparent analysis for the users**.
- Implemented **Daytona sandbox with PostgreSQL and Redis** to provide workspace-based experience with persistent conversations, subagent task dispatch, and interactive UI so users can stop and resume chats smoothly.
- Used **LangGraph, MCP servers, and 30+ tools for market data, SEC filings, search, and web crawling** to feed the agent with the latest market data from various sources.
- Wired **LiteLLM** with support for OpenAI, Anthropic, Gemini, and DeepSeek so the system can switch providers for cost, latency, or compliance in the same agent interface.
- The frontend parses **SSE streams** from the backend and progressively renders model output, file updates, and tool results for a responsive chat experience during long agent runs.
- Used **Axios** with a configurable base URL, **MSW** for development mocking, and **React Router** for clean navigation so the app works cleanly with both local and production backends.
- Managed production lifecycle of a live service (**1000+ users, 11000 daily API calls**), supporting continuous feature release while maintaining system reliability.

Cornell Autonomous Underwater Vehicle Project Team — Ithaca, NY

Software Engineer — Oct 2025 – Present

- Designed an **automated data pipeline** to collect and preprocess pool test data and trained **non-linear ML models** to estimate parameters in non-linear control systems, reducing control error by **80%**.
- Optimized robustness via iterative PID and model-based tuning in varying underwater conditions.
- Presented technical updates in weekly full team engineering meetings.

Projects

<<GroceryManager — Online Grocery Lists Management System Powered by Spring Boot>>

- Designed and implemented **RESTful APIs** using **Spring Boot** with layered architecture and **dependency injection** and improved scalability consistency through **cloud-native** infrastructure design.
- Integrated **PostgreSQL (AWS RDS)** and deployed containerized **Docker** image on **AWS App Runner**.
- Implemented secure **session-based authentication** using **Spring Security**.

<<Pixel Social — AI-Powered Social Platform For Sharing AI Artworks>>

- Implemented **JWT-based authentication** and secure route protection using **React Router**.
- Integrated **DALL·E 3** for AI-generated content and built distributed search infrastructure with **ElasticSearch**.
- Deployed **Golang** backend on **Google App Engine**.

<<MiniSpotify — Lightweight Android Music Streaming Application>>

- Developed Android app using **Kotlin, Jetpack Compose, and MVVM architecture**.
- Implemented local persistence with **Room Databas**e and network layer with **Retrofit**.
- Integrated Google **ExoPlayer** for global audio playback management.
- Applied **Hilt Dependency Injection** to modularize app components and improve maintainability.

<<PaperChat — Online ChatBot Platform Powered by Full-Stack RAG & Tool-Augmentation>>

- Built a **RAG pipeline using LangChain and GPT**, enabling context-aware document reasoning.
- Integrated external web search tools via **MCP and SerpAPI** to enhance retrieval coverage.
- Developed backend services with **Node.js and Express** and built responsive UI using **React and Ant Design**.
- **Reduced hallucination risk** by combining retrieval information with tool-verified responses.